

SET 16- Customer Demographics Analysis

16A. Line Chart

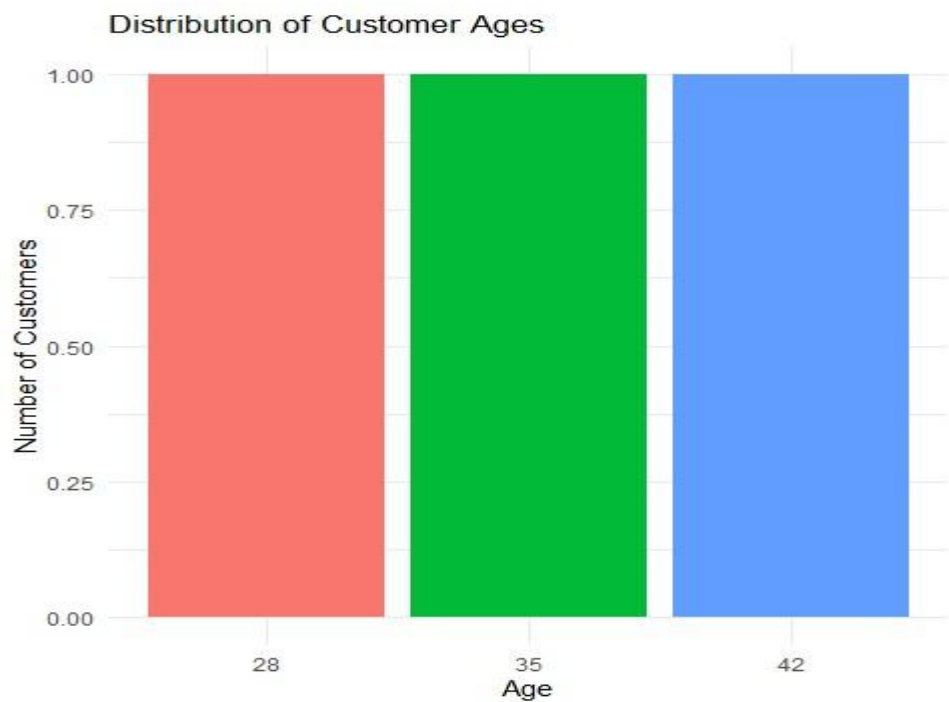
Program:

```
library(ggplot2)

customer_data <-
data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender =
c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
geom_bar() +
  labs( title = "Distribution of
Customer Ages", x = "Age",
y = "Number of Customers"
) +
  theme_minimal() +
theme(legend.position = "none")
```

Output:



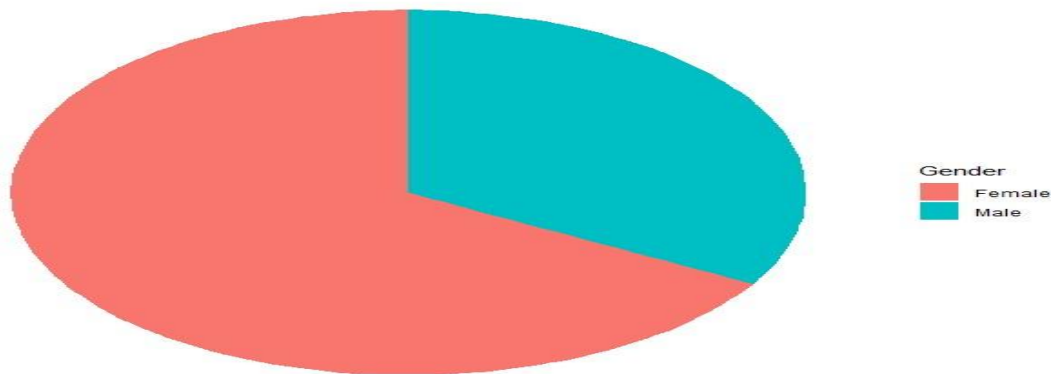
16B. Bar Chart

Program:

```
library(ggplot2)
customer_data <-
data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender =
c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
gender_data <-
as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender","Count")
ggplot(gender_data, aes(x = "", y = Count, fill =
Gender)) + geom_bar(stat = "identity", width = 1)
+ coord_polar(theta = "y") + labs(title =
"Distribution of Customers by Gender") +
theme_void()
```

Output:

Distribution of Customers by Gender



16C. Table

Program:

```
View(customer_data)
```

Output:

R Data: customer_data				
	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000